

CSE7302 - INTERNSHIP

Review-4 Presentation

A Unit of Test Yantra Software Solutions(Robowaves)

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Content

- **About Company or Organization**
- **Working domain or the technology**
- **About your team and reporting Manager**
- **Challenges Faced in Internship**
- **Objectives of the work**
- **Literature Review**
- **Proposed System**
- **Problem Statement**
- **System Requirements**
- **Advantages of Proposed System/Work**
- **System Design**
- **System Implementation**
- **System Testing**
- **Conclusion**
- **References**
- **Github Link**



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About Company or Organization

- **Test Yantra Software Solutions (India) Pvt. Ltd.** is a leading global software testing, quality engineering, and technology services company.
- Founded in **2004**, the company is headquartered in **Bangalore, India**, and serves clients across multiple countries.
- Test Yantra focuses on delivering high-quality IT solutions by combining domain expertise, innovation, and industry best practices.



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About Company or Organization

Test Yantra offers a wide range of IT and quality engineering services, including:

- Software Testing & Quality Assurance
- Automation Testing & Performance Testing
- Digital Transformation Services
- Application Development & Maintenance
- Corporate Training & Skill Development

Working domain or the technology

Domain : Web Technology

Technologies:

- HTML
- CSS
- JavaScript
- ReactJS

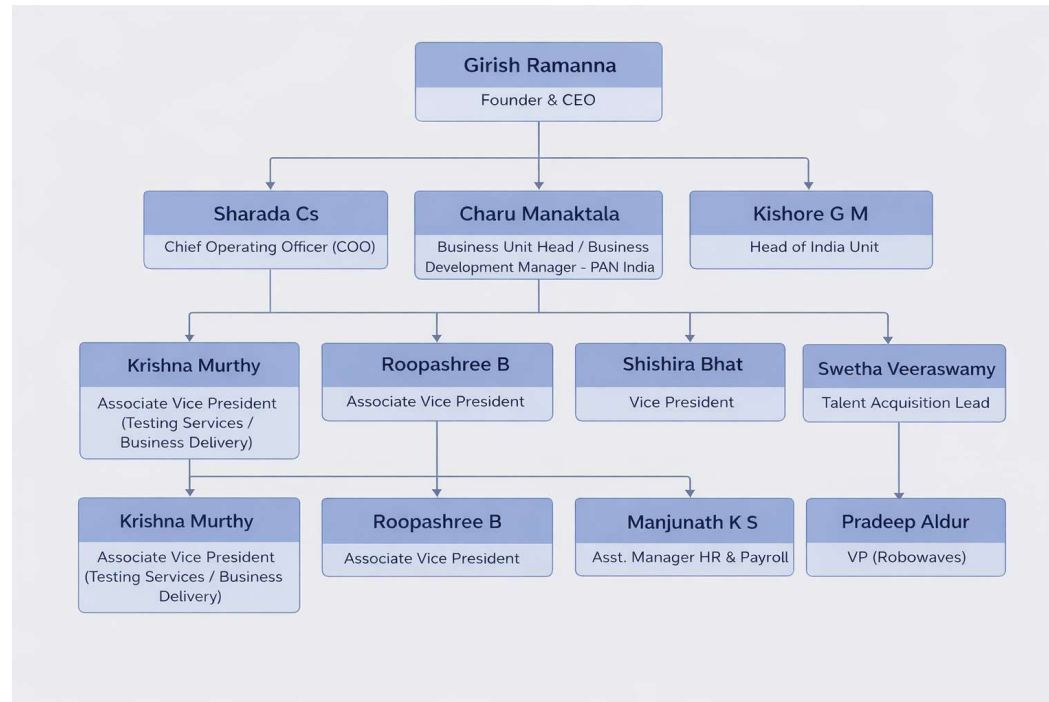


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About your team and reporting Manager



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Challenges Faced in Internship

- Searching for Internship
- Cracking the Interview
- Adapting to the Corporate Environment
- Understanding Real-Time Projects
- Team Coordination



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Objectives of the work

- To develop a **mobile & web-based platform (Namma Raitha)** for direct farmer–retailer interaction
- To provide **fair and transparent pricing** by eliminating middlemen
- To deliver **real-time market price information** for better decision-making
- To integrate **AI-based crop recommendation and disease detection**
- To offer **weather and soil-based insights** for smart farming
- To ensure **multilingual accessibility** for farmers from different regions
- To improve **farmer income, productivity, and efficiency**



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Literature Review

- Agriculture faces challenges like **middlemen, unfair pricing, and lack of market access**
- Many researchers proposed **mobile-based agricultural platforms** to solve these issues
- These systems aim to:
 - Connect **farmers directly with buyers**
 - Provide **market price information**
 - Improve **transparency and communication**
- However, most existing solutions focus on **limited functionalities only**



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Literature Review

Existing Systems & Contributions

- **Content:**
- **Kishan et al. (2025)**
→ Direct farmer–retailer connection, crop listing
- **Bhavani et al. (2025)**
→ Real-time market price visibility
- **Tati et al. (2025)**
→ Includes transport & fertilizer services
- **AgroTIC (2023)**
→ Crop monitoring + marketplace integration
- **Vyas et al. (2024)**
→ Built using Flutter & Firebase (modern tech)
- **Kumar et al. (2020)**
→ Regional language support for accessibility

Proposed System / Work

- Develop a **mobile & web-based platform (Namma Raitha)** connecting farmers directly with retailers
- Eliminate **middlemen** to ensure **fair and transparent pricing**
- Provide **real-time market price data** using government APIs
- Integrate **AI-based crop recommendation & disease detection**
- Enable **multilingual support** (English, Kannada, Tamil, Telugu, Hindi)
- Ensure smooth interaction using **React Native + FastAPI + Supabase**



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Proposed System / Work

- **Core Features:**
- **Direct Market Access**
 - Farmers sell directly to retailers
- **Dynamic Pricing System**
 - Real-time price updates
- **AI-Based Disease Detection**
 - Image-based crop disease prediction
- **Soil-Based Crop Recommendation**
 - Based on N, P, K, pH values
- **Weather Integration**
 - Helps in better farming decisions
- **Negotiation & Order Tracking**
 - Farmers and retailers can negotiate prices

Problem Statement

Agriculture sector suffers from:

- Price exploitation by intermediaries
- Lack of transparency in market pricing
- Limited digital awareness among farmers

Around **55% of farmers lack knowledge of daily market prices**

No unified system providing:

- Real-time pricing
- AI-based insights
- Weather data
- Crop recommendations



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System Requirements

- The system requirements define the **hardware, software, and functional needs** for the Namma Raitha application.
- Supports **farmer–retailer interaction**
- Enables **real-time data processing**
- Integrates **AI models and external APIs**
- Ensures **secure and scalable performance**



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System Requirements

Hardware Requirements:

- Smartphone with camera (for farmers)
- Laptop/PC (for development & admin)
- Stable internet connection
- Cloud server (for backend & database)

System Requirements

Software Requirements

- **Frontend:** React Native (Expo)
- **Backend:** FastAPI (Python)
- **Database:** Supabase (PostgreSQL)
- **APIs:**
 - Market Price API (data.gov.in)
 - Weather API (OpenWeatherMap)
- **Tools:** VS Code, Postman, GitHub

Advantages of Proposed System/Work

- **Direct Market Access**
 - Farmers can sell directly to retailers, eliminating middlemen
- **Fair & Transparent Pricing**
 - Real-time market data ensures better profit for farmers
- **AI-Based Decision Support**
 - Crop recommendation and disease detection improve productivity
- **Weather & Soil Insights**
 - Helps farmers make informed farming decisions
- **Multilingual Support**
 - Accessible to farmers from different regions
- **Data-Driven Farming**
 - Improves efficiency and reduces losses
- **Time & Cost Efficient**
 - Reduces manual effort and unnecessary expenses



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System Design

Frontend (User Interface)

- Built using **React Native**
- Interfaces for **Farmers & Retailers**
- Features: crop upload, price view, order tracking

Backend (Server)

- Developed using **FastAPI**
- Handles **API requests, business logic, and data processing**
- Connects frontend with database and AI models
- **Database & Storage**

Supabase (PostgreSQL)

- Stores user data, crop details, orders, and images
- Provides **authentication & real-time updates**

AI & External APIs

- **CNN Model** for disease detection
- **Market Price API** for real-time pricing
- **Weather API** for weather insights

Workflow

- Farmer uploads crop → Backend processes data



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System Implementation

Development Technologies

- **Frontend:** React Native (Expo)
- **Backend:** FastAPI (Python)
- **Database:** Supabase (PostgreSQL)
- **Tools:** VS Code, Postman, GitHub

AI Model Implementation

- CNN-based **plant disease detection model**
- Trained using **plant disease dataset (Kaggle)**
- Fast prediction (~0.8 sec per image)

API Integration

- **Market Price API** → real-time crop prices
- **Weather API** → weather forecasting
- Backend integrates APIs and sends data to frontend

System Workflow

- Farmer uploads crop details & image
- Backend processes input & calls APIs



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System Testing

- **Types of Testing**
- **Unit Testing**
 - Individual modules tested (price prediction, disease detection, APIs)
- **Integration Testing**
 - Ensures smooth interaction between frontend, backend, and database
- **System Testing**
 - Complete system tested for performance and functionality
- **Acceptance Testing**
 - Verified system meets user requirements (farmers & retailers)



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System Testing

Testing Parameters

- API response time (<250ms)
- Accuracy of disease detection (~98%)
- Correct price recommendations
- Proper API data fetching (weather & market)

Conclusion

- The **Namma Raitha system** provides a digital solution to connect farmers directly with retailers
- Eliminates **middlemen**, ensuring **fair and transparent pricing**
- Integrates **AI-based disease detection and crop recommendation** for better decision-making
- Provides **real-time market prices and weather insights**
- Improves **farmer income, productivity, and efficiency**
- Supports **multilingual access**, making it user-friendly



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Github Link

- <https://github.com/Thanush554/FINAL-YEAR-INTENSHP>



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Q&A



Thank you !!



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